

Dell Technologies AI Fabrics Overview: NVIDIA Spectrum with Dell SONiC

White Paper

H04658

Abstract

This white paper describes NVIDIA Spectrum with Dell SONiC (Software for Open Networking in the Cloud) and provides a general view of the different AI Ethernet Fabrics using open standards networking concepts.

Notes, cautions, and warnings

 **NOTE:** A NOTE indicates important information that helps you make better use of your product.

 **CAUTION:** A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

 **WARNING:** A WARNING indicates a potential for property damage, personal injury, or death.

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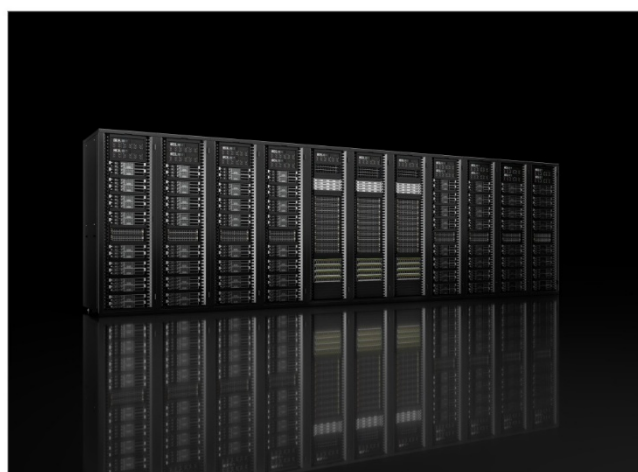
Introduction

Business challenge

Dell Technologies continues to expand its Artificial Intelligence (AI) infrastructure offerings with the release of high-performance NVIDIA Spectrum Ethernet switches with Dell SONiC for Data Center and Cloud environments. The portfolio now includes Dell PowerSwitch SN5600, SN5610, and SN2201. AI solutions are inherently complex, requiring seamless integration of compute, storage, and networking components. Leveraging a familiar and open network operating system can significantly simplify this process, ensuring smoother deployment and faster time-to-value. As AI applications scale to handle massive datasets and growing user demands, maintaining consistent performance becomes a critical challenge. Enterprises need the right infrastructure and ecosystem to support these resource-intensive workloads. AI workloads require lossless, high-throughput data transmission, which can drive up infrastructure and operational costs. Organizations must therefore focus on streamlining, managing, and scaling resources to minimize costs while maintaining operational efficiency. Dell offers a comprehensive product portfolio to deliver a complete AI stack—encompassing networking, compute, storage, and application layers—that is designed to meet these challenges head-on.

Dell SONiC running on NVIDIA Spectrum switches provides an open, flexible, and high-performance networking solution that is tailored for AI workloads. SONiC stands out for the following features:

- **Lossless Ethernet for AI fabrics:** NVIDIA Spectrum with Dell SONiC switches deliver RDMA over Converged Ethernet (RoCE) capabilities, ensuring ultra-low latency and lossless data transmission—critical for AI training and inferencing clusters.
- **Open and familiar operating system:** SONiC offers a Linux-based, containerized architecture that is widely adopted in cloud and enterprise environments. SONiC's openness and modularity simplify integration with automation frameworks and orchestration tools.
- **Scalability and performance:** Spectrum with Dell SONiC switches provides predictable, wire-speed performance with advanced congestion control, enabling AI clusters to scale efficiently without bottlenecks.
- **Ecosystem integration:** SONiC on Spectrum aligns with modern AI infrastructure requirements, supporting open APIs, telemetry, and programmability for streamlined operations and cost optimization.
- **Future-proof architecture:** Combining SONiC's flexibility with NVIDIA's cutting-edge hardware ensures readiness for evolving AI workloads and next-generation networking demands.



AI Cloud
Multi-tenant | Variety of workloads | Ethernet network

Figure 1. NVIDIA Spectrum with Dell SONiC

Document purpose

This white paper provides general guidelines for building a high-performance Ethernet connectivity fabric that is suitable for training and inferencing AI and Generative AI (GenAI) models.

The paper presents the design challenges and fabric recommendations for supporting clusters of GPU-enabled hosts working cooperatively to process new AI and GenAI workloads. These workloads have moved from the realm of scientific and specialized applications and exploded into the general enterprise market.

Creating an infrastructure for AI and GenAI requires careful consideration. This paper highlights the cluster architecture, with a focus on the unique fabric implementations that can mean the difference between a highly successful cluster and one that fails to deliver results. This architecture begins with an understanding of the dramatic pressure that GenAI training workloads place on the network. These jobs require large numbers of GPUs with extensive memory iteratively sharing model data with all the other GPUs until an optimal solution is achieved.

This process can take days or even weeks, and each iteration moves all the data through the fabric in large, synchronized “elephant flows” that must be completed before the next iteration can begin. Elephant flows are large, long-lived network traffic streams that consume high bandwidth. Any delay in any of these transfers is known as “tail latency” about the job completion time. Tail latency can result in the entire cluster being kept idle and not performing its function. Further, given that every GPU is talking to every other GPU, the fabric must be able to spread traffic evenly across the fabric and avoid flow collisions.

Finally, any loss of data or need for retransmission can have dramatic consequences for tail latency and should be avoided. Advanced methods for congestion management and lossless data communication techniques address this issue.

This document describes basic networking concepts and approaches for greenfield AI Ethernet fabric deployments using NVIDIA Spectrum with Dell SONiC.

The document also describes the software and hardware components that are required to build a Dell AI environment based on the Dell PowerEdge XE9680 compute node, explaining how the XE9680 node is paired with the Dell PowerSwitch SN-Series 800 GbE and 400 GbE data center product family running the Dell SONiC operating system.

The document is not intended to imply a preference for one fabric deployment approach over another; these approaches have different merits depending on customer requirements. For detailed analysis and in-depth reviews of each approach, consult the documentation that is available from the relevant solution provider.

Revision history

Table 1. Revision History

Publication date	Part number	Description
March 2026	H04658	Initial release

Disclaimer

The white paper does not provide performance data or sample infrastructure configurations.

Audience

This white paper is intended for AI solution architects, data engineers, IT infrastructure managers, and IT personnel who want to implement AI deployments.

Terminology

The following table provides definitions for some of the terms that are used in this document:

Table 2. Terminology

Term	Definition
AI	Artificial Intelligence
ARS	Adaptive Routing and Switching
BGP	Border Gateway Protocol
Dell SONiC	Dell SONiC implementation
DPU	Data Processing Unit
Elephant flows	Large, long-lived network traffic streams that consume high bandwidth.
ECMP	Equal-Cost Multi-Path
ECN	Explicit Congestion Notification
EVPN	Ethernet VPN
ETS	Enhanced Transmission Selection
GbE	Gigabit Ethernet
GPU	Graphics Processing Unit
NVIDIA Spectrum	NVIDIA Spectrum based hardware Ethernet switches
OOB	Out of Band management network
PFC	Priority Flow Control
RDMA	Remote Direct Memory Access
RoCE	RDMA over Converged Ethernet
SONiC	Software for Open Networking in the Cloud
SuperNIC	A high-performance network accelerator designed for hyperscale AI workloads, providing ultra-fast GPU-to-GPU connectivity using technologies like RDMA over Ethernet.
ToR	Top of Rack
VLAN	Virtual Local Area Network
VXLAN	Virtual Extensible LAN

We value your feedback

Dell Technologies and the authors of this document welcome your feedback on the solution and the solution documentation. Contact the Dell Technologies Solutions team by [email](#).

AI and Dell Technologies

This chapter introduces the AI infrastructure offerings from Dell Technologies, highlighting Dell's comprehensive approach to providing end-to-end solutions for AI deployments.

AI fabric requirements

An AI fabric must be:

- Lossless
- High performance
- Scalable

Large Language Models (LLMs) require the exchange of a large volume of data, creating significant network traffic. Characteristics of AI data traffic patterns are:

- High-rate data exchange taking place during the initial training phase of any model
- Latency-sensitive applications exchanging vast amounts of data in a feedback loop fashion
- Predictable but diverse traffic patterns
- Heterogenous traffic size flows (such as elephant flows)
- Chaotic and bursty dataset patterns

Fabrics for GenAI compute clusters must deliver the highest bandwidth and lowest latency data transfer while avoiding packet loss or any kind of retransmission delays.

Finally, effective utilization of the compute resources depends on minimizing delays that are caused by the network transfers to allow the parallel compute jobs to progress in a synchronized fashion.

Dell networking software

With its complete product portfolio, Dell Technologies has the software and hardware component stack to create and deliver the necessary infrastructure and surrounding components for an AI environment.

Dell SONiC network operating system

Dell Technologies has created a world-class, enterprise-grade version of the open-source SONiC stack. This stack delivers scalable, high-performance, multi-tenancy connections for Hyper-Converged Infrastructure (HCI), Converged Infrastructure (CI), and AI workloads. For AI workloads, this includes the backend fabric, frontend fabric, and Out-of-Band (OOB) network.

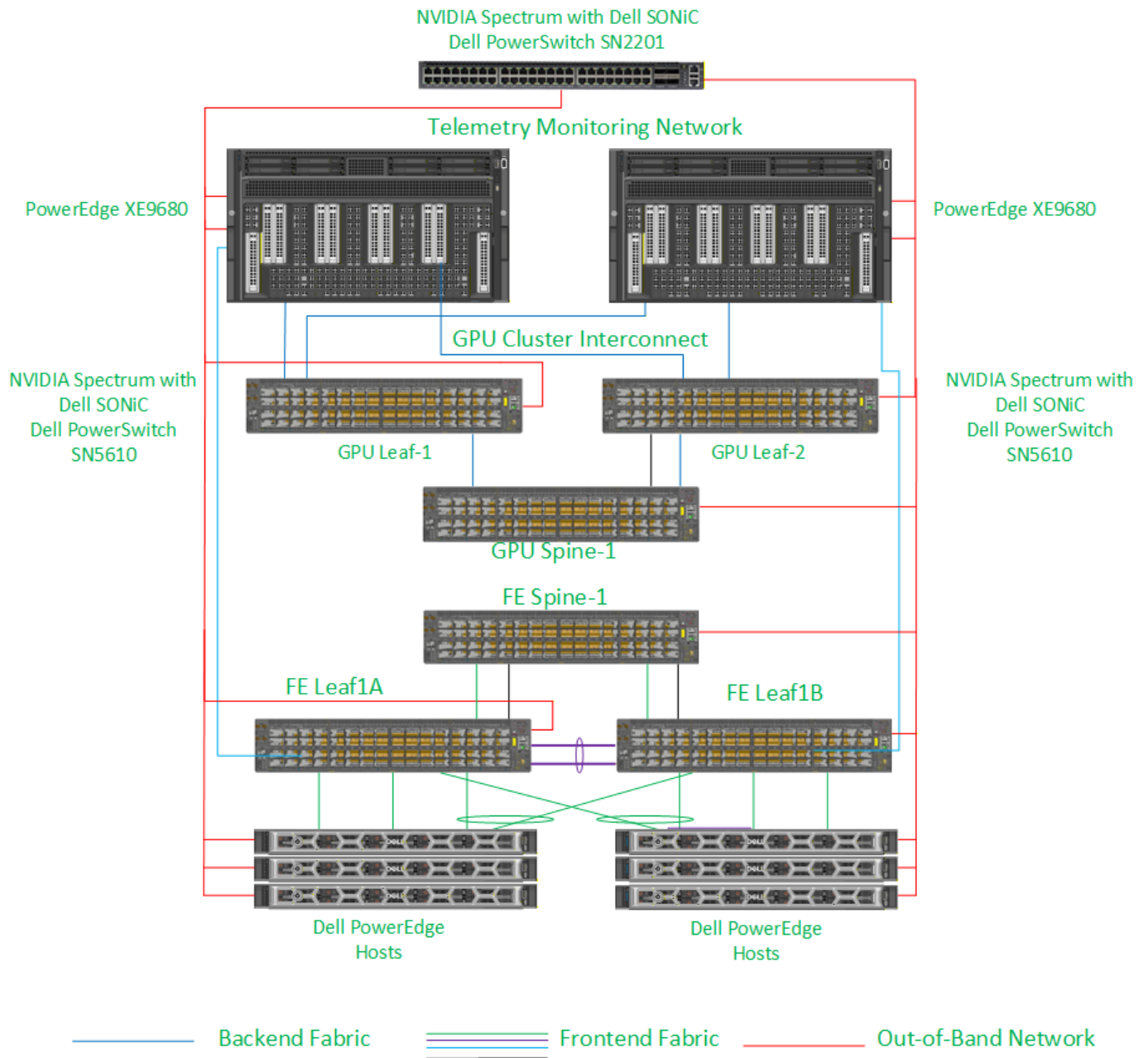


Figure 2. AI fabrics with NVIDIA Spectrum Dell SONiC

NVIDIA Spectrum with Dell SONiC has implemented a series of networking features that facilitate the deployment of a single-rack to multi-rack GPU cluster.

The following figure shows the initial Dell SONiC for AI networking feature set. Each feature aims to address performance, quality-of-service, and ease of management for an AI deployment.

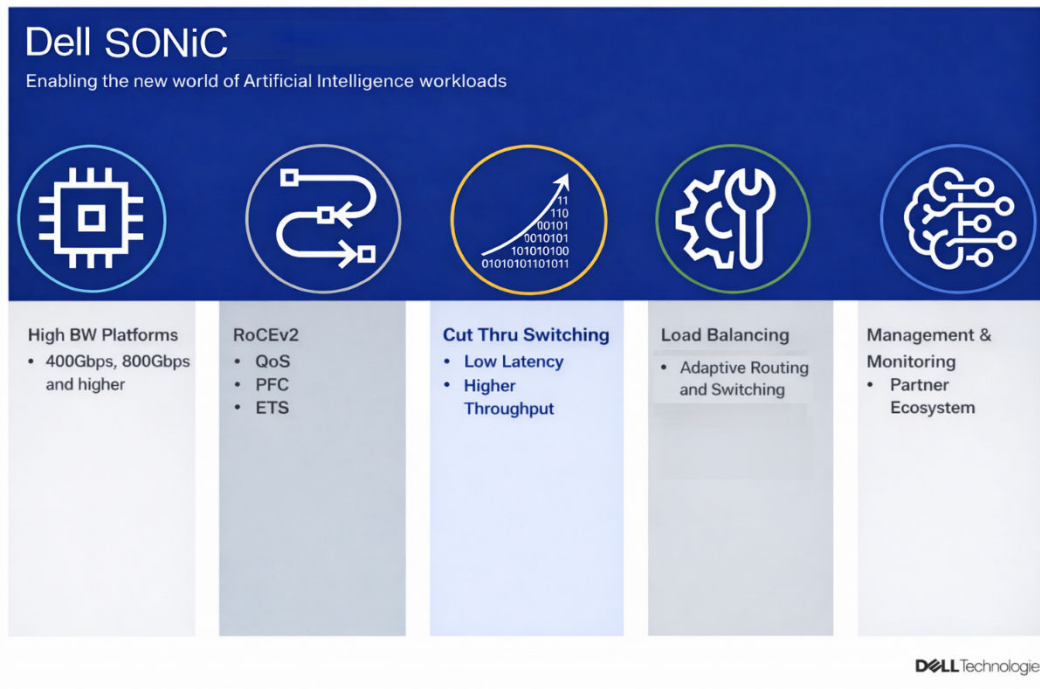


Figure 3. Dell SONiC AI feature set

To address the high-performance requirements of an AI workload, Dell networking provides a switch product portfolio with 400 GbE and 800 GbE switching capacity.

RoCEv2 guarantees reliable performance for complex and demanding AI traffic patterns, even in congested network fabrics. AI traffic can be classified and marked as “strict priority” traffic in the fabric. The classification as a strict priority ensures that AI traffic is assigned to a specific output queue that is serviced before any other traffic in the fabric. This classification ensures smooth data switching from end-to-end, in this case from GPU-to-GPU.

Finally, load balancing features such as Adaptive Routing and Switching (ARS) address points of congestion to uniformly distribute AI traffic across multiple 400 GbE ports.

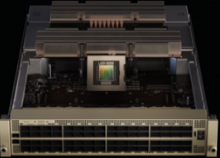
Together, these features create a solid foundation on which to create a lossless, high-performance, and scalable Ethernet fabric.

Dell networking hardware

The networking hardware components of the Dell AI fabric solution design are purpose-built for the AI world. The hardware product family is high-performing, efficient, and based on open standards. These characteristics make a Dell end-to-end solution design a good fit for unique workloads that require efficient, high-performing, and easy-to-manage deployments.

Dell Ethernet: NVIDIA Spectrum with Dell SONiC

The Dell PowerSwitch portfolio has been expanded to provide an extensive vendor hardware choice for building an infrastructure for AI solutions. In this document, the Dell PowerSwitch SN5600, SN5610, and SN2201 switches are the primary 800 GbE and 100/1 GbE switches respectively. With the Dell PowerSwitch SN5600 and SN5610 Spectrum-4 switches providing up to 51.2 terabits per second (Tbps) of switching capacity, these three models are ideal for any kind of AI workload.



Spectrum SN5610 Ethernet Switch



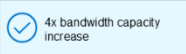
BlueField-3 SuperNIC



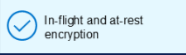
ConnectX-7 SuperNIC

Purpose-Built Ethernet Fabric for AI

Dell PowerSwitch SN5610









Source: NVIDIA comparison between traditional Ethernet (Spectrum with our best RoCE vs. Spectrum-X features turned on. 1.6x refers to the better bandwidth utilization achieved with adaptive routing.

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Figure 4. NVIDIA Spectrum with Dell SONiC, Dell PowerSwitch SN5610

Dell PowerSwitch NVIDIA SN2201

Ethernet Switch Out of Band Network



48x Base-T + 4x QSFP28 100 GbE

Powered by the Spectrum ASIC

High performance combined with feature-rich layer-2 and layer-3 forwarding

*Source: NVIDIA. All the performance comparisons are comparing against previous versions of NVIDIA switches.

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Figure 5. NVIDIA Spectrum with Dell SONiC, Dell PowerSwitch SN2201

In this document, the NVIDIA Spectrum with Dell SONiC PowerSwitch SN2201 is used as the OOB management switch. The SN2201 OOB fabric connections are 1 GbE copper-based RJ45 connections.

Networking Guide and Requirements

AI Ethernet fabric requirements

The AI Ethernet fabric must provide certain key features and leverage open standards to become a compelling fabric interconnect of choice for the AI world.

Standards

The required standards include:

- Interoperable:** The Dell AI fabric must operate at the highest level and leverage a well-established network ecosystem that is based on proven open standards such as Ethernet. A flexible architecture can be achieved with an Ethernet-based approach.
- High-performance:** AI workloads are unique in that they require specific network or fabric properties to perform optimally.

Features

With Dell SONiC as the networking operating system, an AI-specific feature set that includes cut-through switching, RoCE version 2, and ARS provides the network performance that supports AI workloads. For more information, see [Dell SONiC network operating system](#). In addition to the Dell SONiC software feature set, NVIDIA Spectrum with Dell SONiC provides the necessary non-blocking 800 GbE switching fabric capacity:

- **Scalable:** With the Dell networking product portfolio, AI fabrics customizations enable a range of GPU cluster deployments. The requirements of AI workloads can range from a single GPU to a cluster of multiple GPUs. This white paper explores some relevant AI fabric topologies: Pure Rail topology and Rail Optimized topology.
- **Lossless fabric:** To support RoCEv2, an Ethernet fabric must leverage PFC to be lossless.
- **ARS:** Dynamic path selection avoids sub-optimal links for enhanced performance by quickly reacting to link failures.
- **Congestion control:** RoCEv2 with PFC and ECN plus telemetry enable end-to-end congestion control.

Building a GPU cluster with the PowerEdge XE9680

Dell PowerEdge XE9680

The PowerEdge XE9680 is a 6U server, Dell's first eight-GPU platform. This server is designed to significantly enhance application performance by driving the most complex AI, machine learning, deep learning (ML/DL), and high-performance computing (HPC) workloads.

The PowerEdge XE9680 features up to two 56-core 4th generation Intel® Xeon® Scalable processors and offers the highest GPU memory capacity and bandwidth currently available, making it capable of managing large and complex models and datasets.

Each GPU in the PowerEdge XE9680 server is connected through PCIe to a dedicated NIC that interfaces with the fabric at 400 GbE speed, as shown in the following figure. Direct attach copper (DAC) and fiber cabling are supported.

Dell PowerEdge XE9680

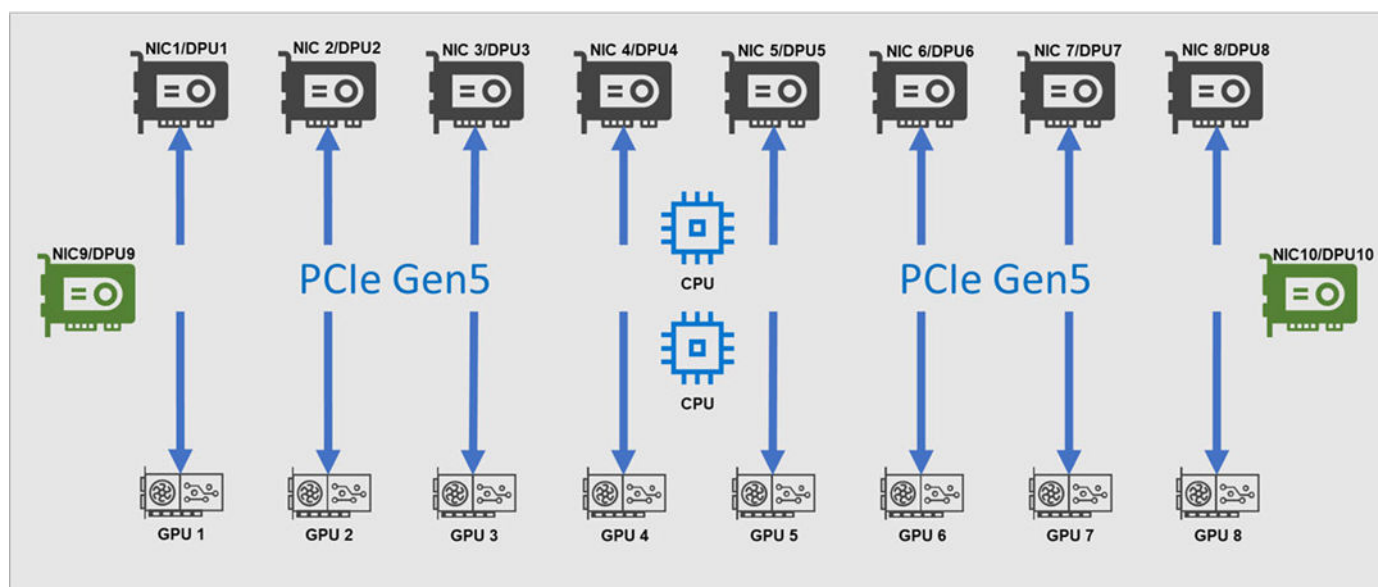


Figure 6. Components of a Dell PowerEdge XE9680 server

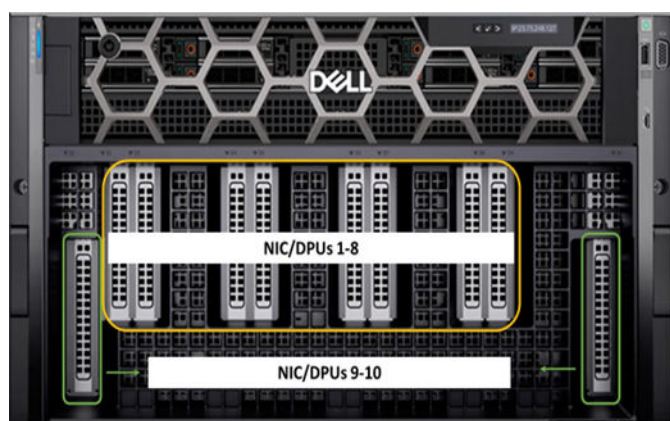


Figure 7. Dell PowerEdge XE9680 – front view

The NICs in the two left-most and right-most PCIe slots in the preceding figure are used for the frontend fabric connections, while the DPU NICs, such as the NVIDIA CX7, CX8, and BlueField-3, are coupled with the eight GPUs and used to connect to the Scale-out Fabric (see Building a GPU cluster).

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The NICs in the two PCIe slots on the far right and far left of the preceding figure are used for the frontend fabric connections, while the eight DPU NICs such as the NVIDIA CX7, CX8, and BlueField-3 are coupled with the eight GPUs and used to connect to the Scale-out Fabric (see Building a GPU cluster).

Building a GPU cluster

The PowerEdge XE9680 server integrates eight GPUs and up to 10 NICs/DPUs. Multiple fabrics are needed to create a GPU cluster based on the XE9680 model, as shown in the following figure:

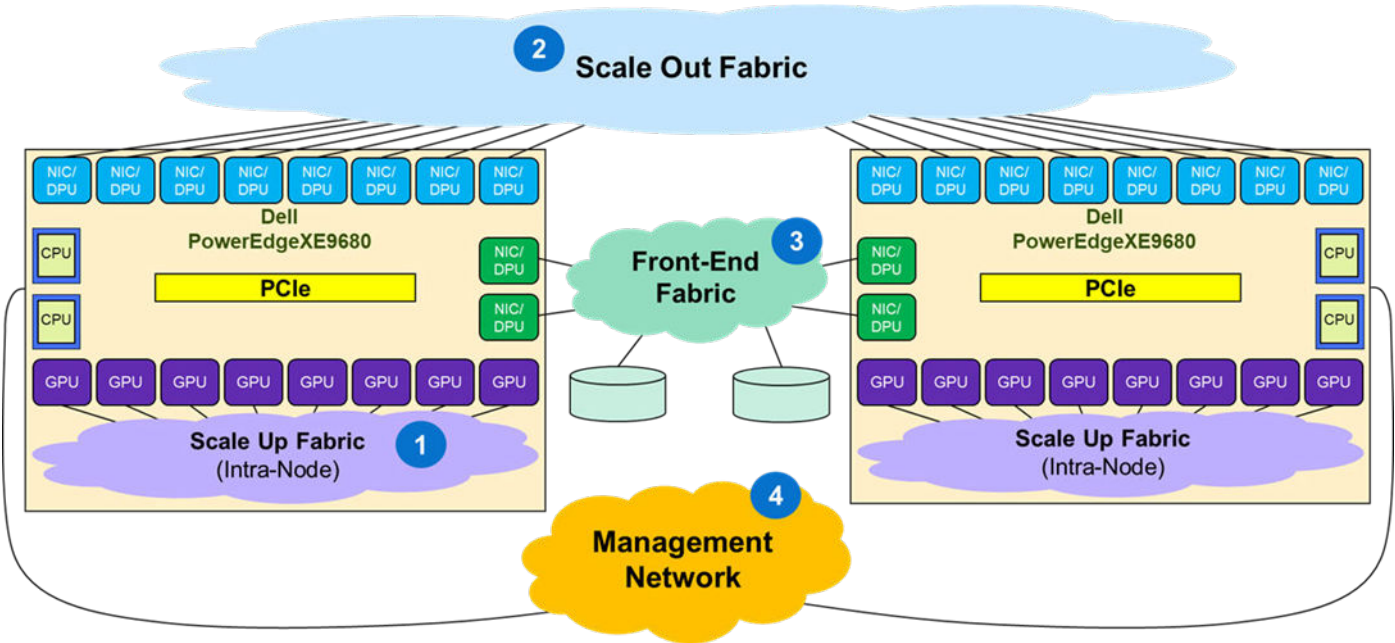


Figure 8. Example of a PowerEdge XE9680-based GPU cluster

The eight GPUs within an XE9680 cluster are interconnected by an intra-node scale-up fabric that creates a high-bandwidth GPU domain. Eight of the NICs/DPUs are connected through PCIe with the eight GPUs and used to exchange parameters among GPUs belonging to different chassis through a scale-out fabric. The remaining two NICs/DPUs are connected to a frontend fabric that supports storage deployment, applications, and in-band cluster management.

The preceding figure shows the different fabrics that support the deployment of a typical AI workload. The label numbers correspond to the following components:

- 1. Scale-up Fabric:** This fabric interconnects the eight GPUs to form a high-bandwidth domain, usually through proprietary network technologies such as NVIDIA NVLink or AMD XGMI.
- 2. Scale-out Fabric:** This fabric is used by the set of GPUs composing a GPU cluster to exchange AI workload parameters.
- 3. Front-end Fabric:** This fabric supports the deployment of the storage, application, and in-band cluster management components of the AI solution.
- 4. Management Network:** This fabric supports the overall environment management network for the solution.

The terms "scale-up fabric," "scale-out fabric," and "front-end fabric" are expected to become standard industry terminology. For reference, the following terms are also in use:

- Scale-up Fabric:** GPU fabric, back-end fabric, front-end fabric
- Scale-out Fabric:** Host fabric, east-west fabric, back-end fabric, front-end fabric
- Front-End Fabric:** Storage/access fabric, north-south fabric

This document focuses on the scale-out fabric, which all GPUs in the cluster use to exchange parameters during the long, intensive training jobs that produce a new model.

The document also considers the key considerations for building a frontend fabric to support storage access and application access for the cluster.

AI fabrics

AI workloads incorporate three fabrics or networks:

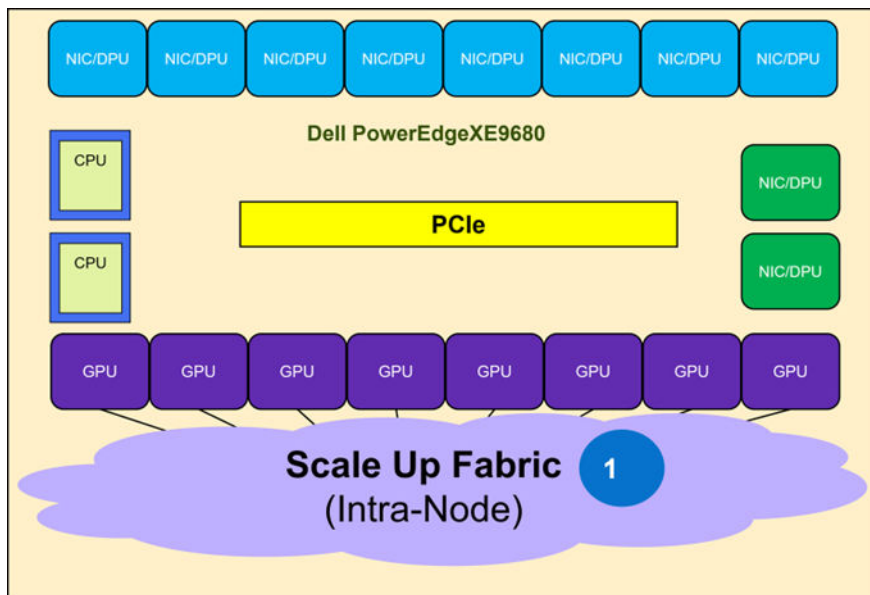
- Backend fabric (scale-out)
- Frontend fabric
- OOB management network

Scale-up fabric

GPUs provide not only numerous compute cores, but also their own communication capabilities. For example, an NVIDIA H100 GPU provides more than 10,000 computation CUDA cores and 18 NVLink4 interfaces, each running at 200 Gb/s, an aggregate bandwidth of 450 GB/s. The most advanced NICs today run at 400 Gb/s, while a single H100 GPU can drive a bandwidth of 450 GB/s, more than an order of magnitude higher. Looking ahead, the GB300 (Blackwell Ultra) has 20,480 CUDA cores and 640 Tensor cores, with 288 GB HBM3e memory. The GB300 uses an NVLink 5 with a bidirectional bandwidth of 1.8 TB/s.

As shown in the following figure, the scale-up fabric interconnects the eight GPUs in an XE9680 to form a high-bandwidth domain using the native interfaces of the GPU. This type of fabric enables efficient sharing and coordination of embedded resources, allowing multiple GPUs to work together. Scale-up fabrics can seamlessly address complex tasks such as AI training and fine-tuning and are transparently supported by the communication library that the GPU vendor provides.

Figure 9. Example of a scale-up fabric cluster



A scale-up fabric:

- Uses different technologies depending on the vendor:
 - NVIDIA uses switched NVLink.
 - AMD uses a full mesh of point-to-point XGMI links.
 - Intel uses Ethernet.
- Provides high bandwidth among the interconnected GPUs, but is spatially limited to a single server.
- Is used to build a GPU cluster that is not limited to a single node.

Scale-out fabric

As shown in the following figure, the scale-out fabric interconnects the NICs that are coupled with the GPUs in an XE9680 server to build a GPU cluster that includes more than a single node. The scale-out fabric is used by the set of GPUs composing a GPU cluster to exchange AI workload parameters.

2 Scale Out Fabric (Inter-Node)

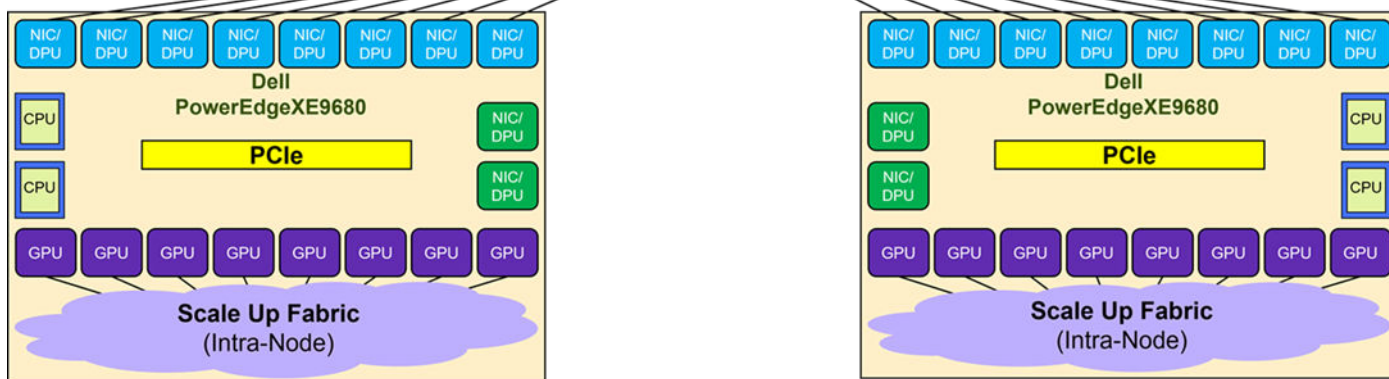


Figure 10. Scale-out fabric cluster example

When deploying a scale-out fabric, take into consideration that:

- EVPN VXLAN eBGP is not deployed as the networking protocol because multitenancy GPU service is not a requirement. Deploying plain BGP as the network protocol in the fabric is recommended.
- Switch redundancy such as MC-LAG is not needed. Link redundancy is achieved by the use of multiple active links between leaf and spine as well as GPU connections to multiple leaf switches.

Keep in mind that a scale-out fabric has several possible topologies. This document reviews two of these topologies:

- Pure Rail topology
- Rail Optimized topology

NOTE: These guidelines do not provide an exhaustive networking design checklist. Fabric configuration requires detailed planning and design throughout the entire deployment cycle.

Pure Rail topology

The following figure shows the connectivity properties of the Pure Rail topology:

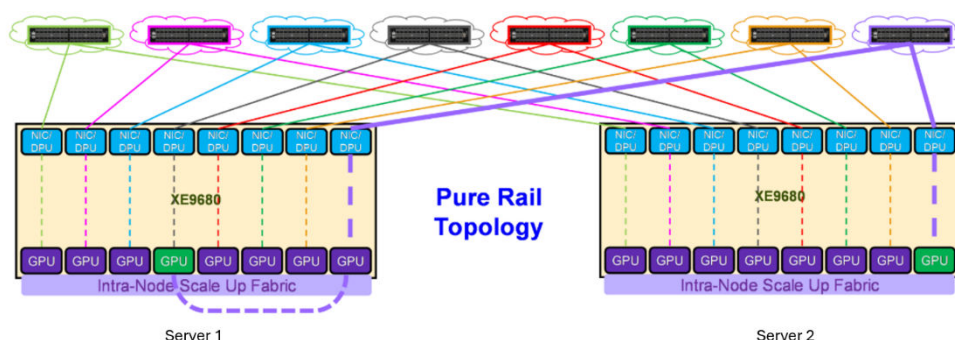


Figure 11. Pure Rail topology

In this topology, each NIC is connected to a different switch (or spine-leaf network) called a rail, as shown. This topology enables direct scale-out fabric communication only between GPUs belonging to the same rail. The scale-up fabric participates in scale-out fabric communications when traffic needs to “cross rails.”

For example, to communicate with GPU 8 on server 2, GPU 4 on server 1 must first transfer its data into the memory of GPU 8 on server 1, known as “crossing the rails.” GPU 8 on server 1 can then communicate through NIC 8 on server 1 with GPU 8 on server 2, which is coupled with NIC 8 on server 2.

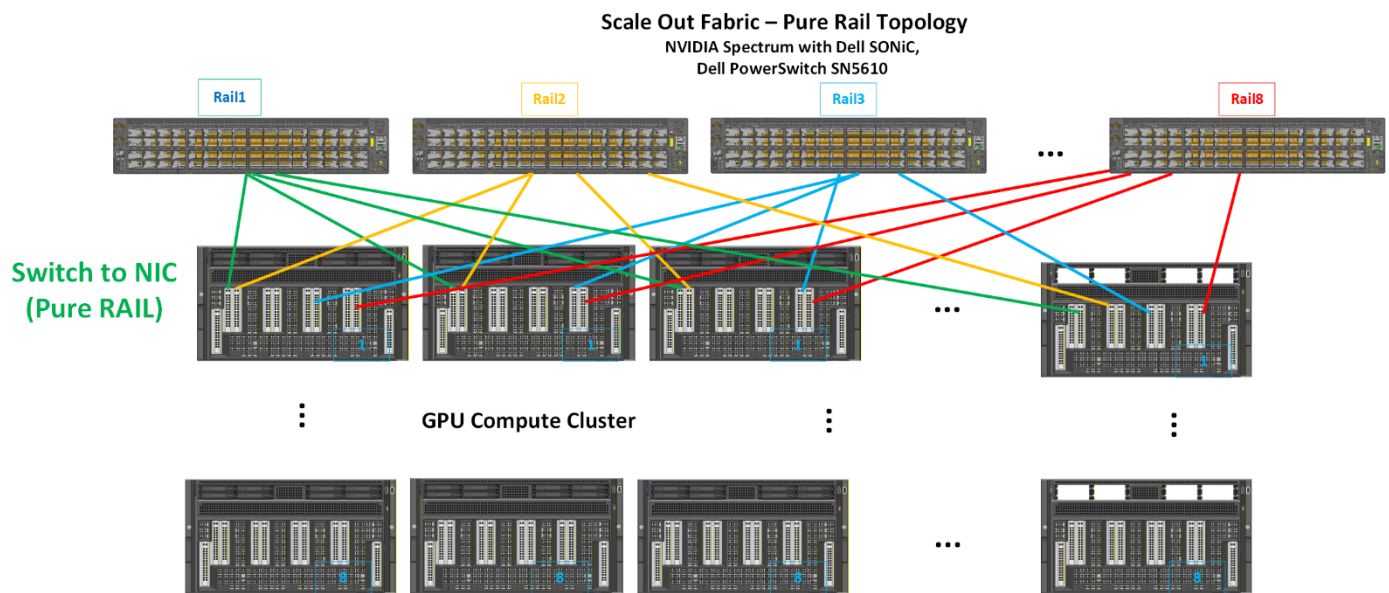


Figure 12. Scale-out Fabric - Pure Rail topology example

The preceding figure shows an eight-rail topology interconnecting eight PODs. Each POD consists of eight XE9680 servers, each of which comes with eight GPUs, creating a fabric size of 512 GPUs. Each GPU from each POD has a dedicated 400 GbE connection. For example, Rail1 is an NVIDIA Spectrum with Dell SONiC SN5600 switch, with 64 x 400 GbE ports. POD_1 consists of eight XE9680 nodes. Each XE9680 node has eight GPUs, resulting in 64 GPUs per POD. Each GPU 1 from each POD connects to Rail1, GPU 2 connects to Rail2, and so on in sequence.

Rail Optimized topology

The following figure shows the connectivity properties of the Rail Optimized topology:

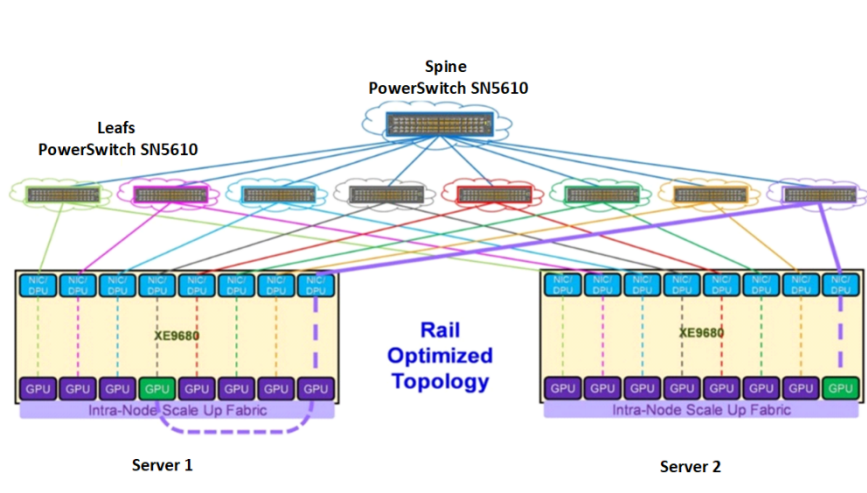


Figure 13. Rail Optimized topology

In this topology, each NIC is connected to a different switch (or spine-leaf network), called a rail. The rails are also interconnected at an upper tier, so this topology provides two ways to cross rails—through the scale-up fabric (preferred) or through the upper tier of the scale-out topology.

To communicate with GPU 8 on server 2, GPU 4 on server 1 can:

- Transfer its data into the memory of GPU 8 on server 1 (cross the rails), and then GPU 8 on server 1 can communicate through NIC 8 on server 1 with GPU 8 on server 2, coupled with NIC 8 on server 2.
- Send its data to NIC 4 on server 1, which can reach through the upper tier NIC 8 on server 2, which is connected to GPU 8 on server 2.

This property allows AI workloads to perform better on a Rail Optimized topology than on a Pure Rail topology, because the current collective communication libraries are not fully optimized for the Pure Rail topology. Rail Optimized is therefore the recommended topology to build a scale-out fabric (see Tradeoffs). In this example, the Dell PowerSwitch SN5610 serves as the baseline switch to build a scale-out fabric using this topology.

The following figure shows an example of a large Rail Optimized topology with the NVIDIA Spectrum with Dell SONiC on Dell PowerSwitch SN5600 or SN5610:

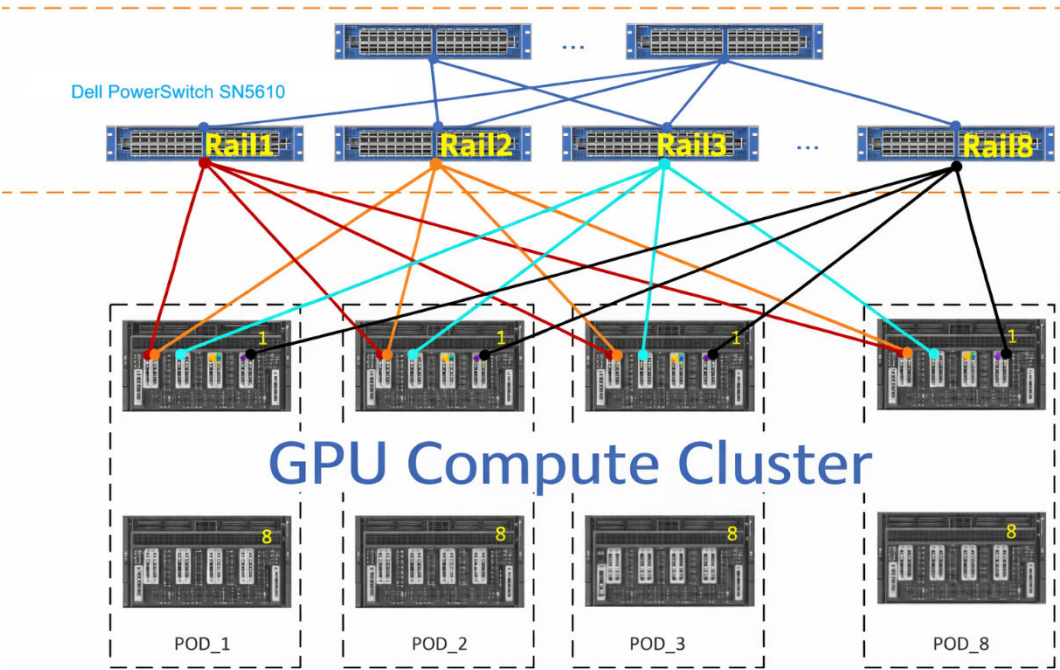


Figure 14. Scale-out fabric - Rail Optimized topology example

The Rail Optimized topology provides the same set of benefits as a Pure Rail topology in addition to scalability and connectivity through the upper tier (spine). At times, the GPU to GPU across different nodes may not take the optimum path (leaf), and the upper tier must provide connectivity between the different rails.

Example: The red GPU on NODE_1 POD_1 wants to communicate with the light blue GPU on NODE_1 POD_3. Ideally, the red GPU would use the internal fabric on NODE_1 to communicate with the light blue GPU on NODE_1, and then the light blue GPU on NODE_1 would transmit this data to Rail3. If for any reason this data path is not available, the red GPU transmits its data upstream towards Rail1, Rail1 transmits forward towards S1, and S1 sends this data downstream to Rail3.

Tradeoffs

The Pure Rail and Rail Optimized topologies provide better GPU reachability, specifically more GPUs at 1-hop distance, for the Dell PowerSwitch SN5610 switches, as shown in the following table. These topologies allow better performances for AI workloads and are therefore recommended over the TOR-wired Clos topology.

Table 3. GPU reachability by topology

Topology	High bandwidth domain	1-hop away	3-hop away
		Dell PowerSwitch SN5610	Dell PowerSwitch SN5610
TOR-Wired Clos	8 GPUs (single node)	64 GPUs (8 nodes)	8,192 GPUs (1,024nodes)
Rail	8 GPUs (single node)	512 GPUs (64 nodes)	65,536 GPUs (8,192 nodes)

Frontend fabric

The frontend fabric is the network infrastructure interconnecting the different functions of the AI environment. This fabric facilitates communication, compute resource sharing, and data transfer among the interconnected clusters in the cluster.

The frontend fabric also offers end users access to the servers and applications that are a necessary part of the AI environment and supports connectivity for network management through in-band interfaces.

When integrated with the storage fabric, the frontend fabric facilitates access to the storage, leading to a comprehensive unified fabric.

The bandwidth link speed from the PowerEdge XE9680 node to the frontend fabric is 400 GbE using NICs 9 and 10 (see Figure 8). Those NICs connect to Dell SONiC switches.

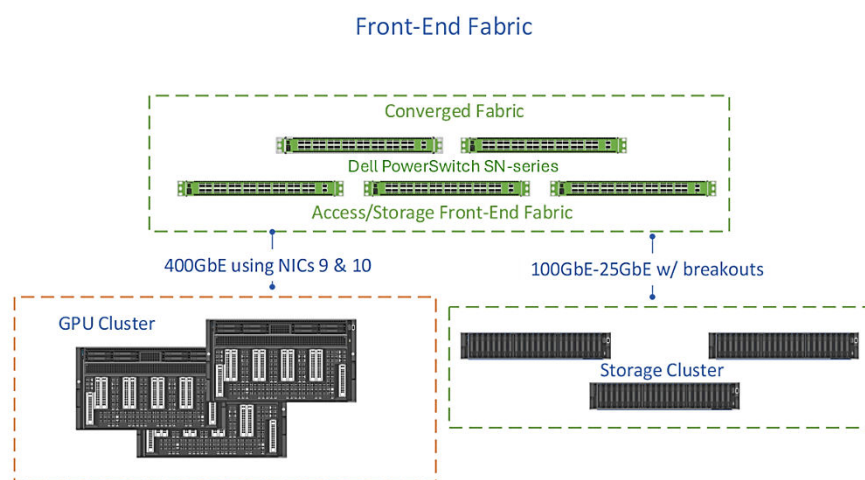


Figure 15. Frontend Fabric

The frontend fabric is a converged fabric where storage, application, and in-band GPU cluster management connections take place at speeds ranging between 400 and 25 GbE.

The frontend fabric uses EVPN VXLAN and standard Layer 2 or Layer 3 to provide services such as multitenancy access into the AI environment or straight access to storage services from the GPUs as needed. These VLAN segments—storage, GPU application, and in-band cluster management—are then routed within the frontend fabric to reach outside the AI environment or stay isolated as a private AI environment sandbox.

In most cases, the connections from the storage, application, and in-band cluster management to the PowerSwitch operate at Layer 2. Each of the different services is assigned a respective VLAN as traffic leaves the devices.

Management network

The management fabric provides out of band (OOB) management connectivity—local or remote—into the AI infrastructure (switches) and into devices such as storage appliances and servers.

The OOB network encompasses simple iDRAC connectivity for the servers, as well as management interfaces for the storage appliances and networking devices. The typical link speed of these connections is 1 GbE using Cat-RJ45 copper cables.

The following figure shows the OOB connections from all the devices in the Dell AI environment:

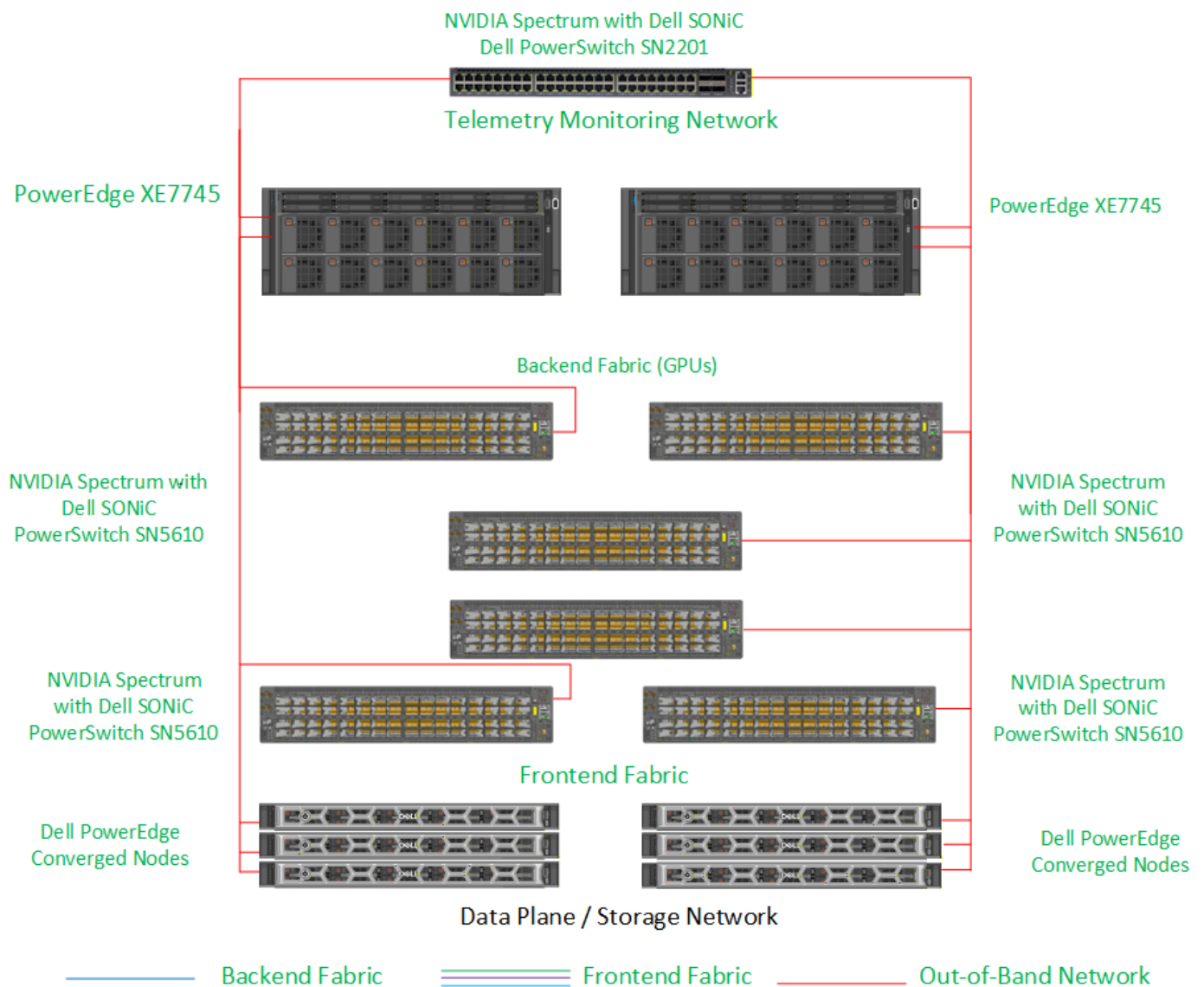


Figure 16. Out-of-Band Management Network

The OOB network uses the NVIDIA Spectrum with Dell SONiC on the Dell PowerSwitch SN2201 switch. This switch supports Layer 2 and Layer 3 switching, and the port speeds are 1 GbE. The Dell PowerSwitch SN2201 switch has four QSFP28 100 GbE ports, which can be used to uplink into an existing network.

When deploying the OOB Ethernet fabric, take the following best practices into account:

- Both fabric types (Layer 2 and Layer 3) are supported on the NVIDIA Spectrum with Dell SONiC PowerSwitch SN2201 switch.
- If the OOB management fabric is based exclusively on Layer 2, all Dell PowerSwitch SN2201 switchports are assigned to a single VLAN or a small number of VLANs. These VLANs are then trunked over the 100 GbE SFP uplink ports on the Dell PowerSwitch SN2201 and transmitted upstream to be routed by a different device.
- Enable spanning-tree whenever Layer 2 is being deployed.
- If the OOB management fabric is based purely on Layer 3, it is likely that the connections from the Dell PowerSwitch SN2201 are Layer 3 point-to-point using a /31 subnet IP address scheme. These IP addresses are then routed throughout the OOB management fabric, providing remote access into the AI environment.

The 10/100GbE SFP uplink ports are still used to connect to an existing network, providing external access into the AI environment.

Dell AI Ethernet Fabric – Orchestration and Monitoring

Introduction

The Dell AI Ethernet fabric consists of two major components:

- Infrastructure or fabric
- Workloads such as GPU, storage, and management clusters

AI infrastructure orchestration and monitoring

Dell offers infrastructure automation and monitoring capabilities through Dell SmartFabric Manager.

SmartFabric Manager for SONiC

SmartFabric Manager (SFM) for SONiC is Dell's answer to the industry demand for simplicity in network management. The SFM for SONiC tool transforms the daunting task of network setup and maintenance into a streamlined, intuitive process. SFM for SONiC automates deployment and management, offering customizable blueprints and an automated fabric discovery that can reduce manual errors, cut operational costs, and enhance network reliability.

SFM for SONiC not only simplifies network design and deployment, but also empowers IT teams with control and visibility. SFM for SONiC optimizes efficiency through standard APIs for seamless automation and provides a detailed performance insight through advanced analytics and monitoring. Customers using SFM for SONiC can reduce configuration errors, deploy with confidence, manage less, innovate more, and enjoy increased network availability and performance.

Dell SmartFabric Manager for SONiC

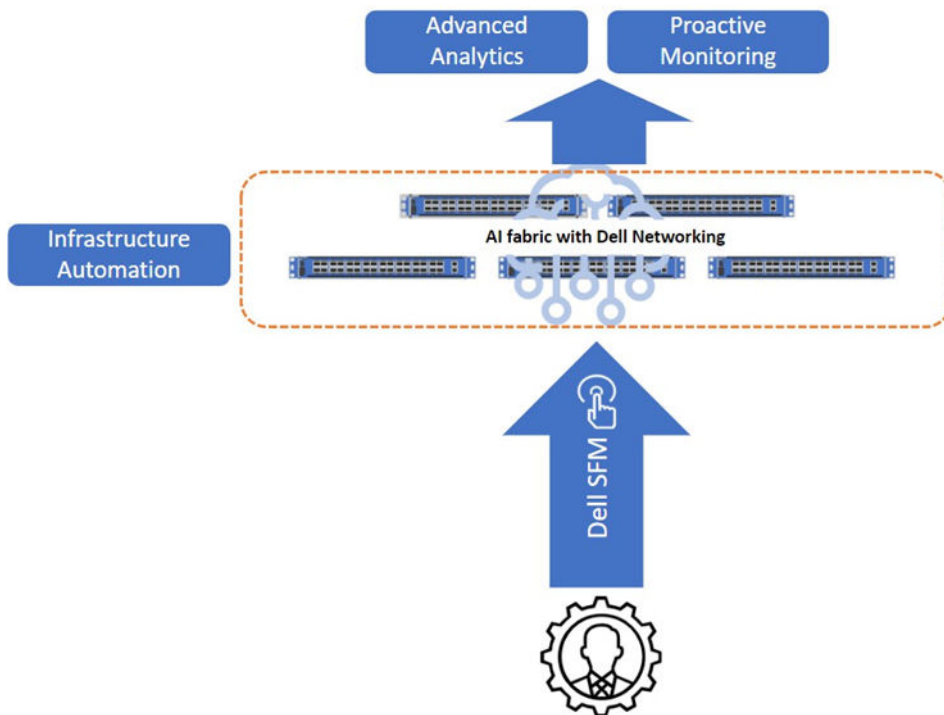


Figure 17. Dell Smart Fabric Manager for SONiC

SFM for SONiC provides:

- **Fabric discovery, validation, and auto configuration:** Wiring detection and simplified network configuration that save time and reduce manual errors by identifying issues quickly and eliminating the need for specialized skill sets.
- **Blueprints for success:** User-defined and predesigned blueprints for Dell solutions and GenAI save time and allow for a 'design once, deploy n times' approach across sites or fabrics.
- **REST API support:** A REST API for all capabilities enables customers to integrate fabric level configurations with scripting.
- **Life cycle Management:** LCM provides uniform NOS version maintenance, planned and scheduled NOS upgrades, auto rollback on failure, and route around in maintenance mode to reduce downtime.
- **Advanced analytics and monitoring:** An advanced dashboard provides a deep-dive view into the fabric, quickly identifying problems and congestion and triggering remedial action to prevent unexpected downtime.

Alongside the embedded orchestration and monitoring capabilities that Dell SONiC provides, a key partnership with BeyondEdge (BE) further enhances this offer.

BE Networks Verity

As an orchestrator, BE Verity can leverage telemetry and metadata to baseline the fabric and detect and respond to spikes in several ways:

- Maintain a desired-state (intent) configuration for each Dell SONiC switch and continuously compare operational state to intended state, alerting on any deviation.
- Detect unauthorized configuration changes on SONiC switches, for example, adding ACL exceptions, modifying VRF/EVPN settings, or disabling PFC/ECN policies outside approved change control.
- Enforce a controlled Zero-Touch Provisioning (ZTP) onboarding process for fabric switches, preventing "snowflake" or rogue devices with ad hoc configurations and ensuring that every AI-fabric switch runs a known NOS image and configuration that is aligned with Dell's secure-supply-chain practices.
- Push only hardened, validated SONiC images from the Verity repository during upgrades, ensuring that switches receive only known-good software builds.
- Lock switches into a "read-only" safety mode through RBAC, blocking malicious or accidental configuration pushes while maintaining full telemetry and observability.

BE Networks Verity: From boxes to a standing network in minutes

Day 0: Design, Build & Turn -up

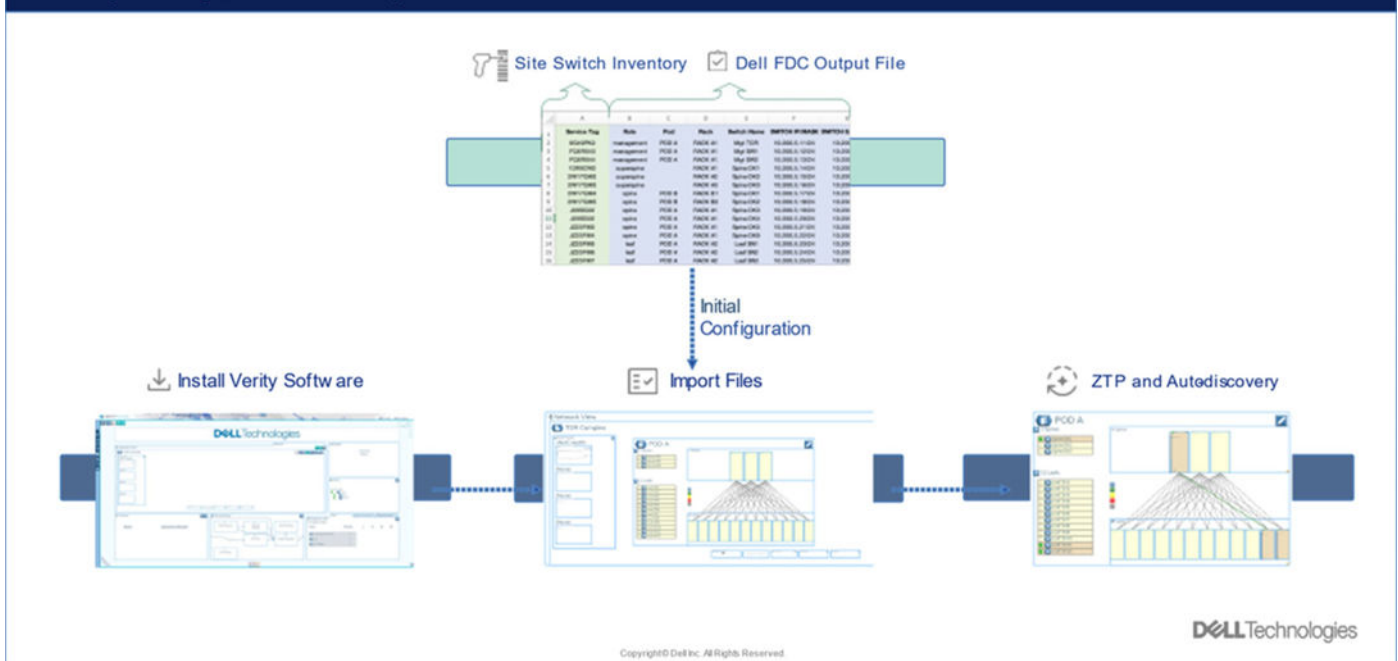


Figure 18. BE Networks Verity – Day 0 process

The preceding figure shows the basic workflow that Verity follows during the automated orchestration of the fabric. The following figure shows the zooming capabilities of Verity. Verity can amplify configurations and events in the fabric.



Figure 19. BE Verity system-level detail view

Dell AI infrastructure orchestration is also deployable by Ansible. Dell SONiC supports the latest Ansible modules and collections for creating YAML playbooks that can automate the infrastructure.

Security

By using Verity, the solution:

- Operates on a Dell-designed AI security architecture that integrates seamlessly with the Dell AI Factory, AI Ethernet fabrics, and the customer's existing security stack.
- Provides strong traffic and user segmentation through EVPN and VRF-based multitenancy, reinforcing segmentation with Verity security best practices.
- Streams SONiC gNMI and telemetry data from the fabric to the Managed Service and Security Operations Center (SOC), enabling continuous, real-time monitoring.
- Uses CrowdStrike Falcon Cloud Security, Next-Gen SIEM, and the Zscaler Eclipse AI firewall to monitor AI prompts, outputs, models, data, and infrastructure 24/7.
- Enables SOC analysts to investigate alerts, perform AI-specific threat hunting (such as data leakage or model abuse), and coordinate response actions with the customer team.

Next steps

State-of-the-art host and GPU connectivity for the industry is based on the RDMA protocol, which can be implemented over Ethernet using RoCE standards. This protocol is supported by high-speed switching, with 800G port speeds as the current benchmark. RoCE is further enhanced by sophisticated load-balancing and congestion-management techniques that ensure lossless, low-latency data transfers to maximize utilization of the compute resources and provide optimal training times and inferencing responses.

AI workloads with NVIDIA Spectrum with Dell SONiC switching options support 800G bandwidth RDMA transmission, and the associated load-balancing and congestion-management techniques that are required to meet GenAI traffic requirements.

Factors that influence the final selection of a fabric technology can include multi-vendor support, open standards, cost and supply constraints, or specific performance and compatibility concerns. The final recommendation must take account of specific customer requirements. For detailed information about the solution options, see the applicable vendor documentation and related white papers on the [Dell Technologies Info Hub for Networking Solutions](#).

Support and References

Technical support

Your first point of contact is [Dell Support](#).

Dell Technologies documentation

The following Dell Technologies documentation provides additional relevant information. Access to these documents depends on your login credentials. If you do not have access to a document, contact your Dell Technologies representative.

- [Dell AI Switches](#)
- [DellPowerEdge XE9680](#)
- [Dell PowerEdge XE7745](#)
- [Dell PowerScale](#)
- [Dell Artificial Intelligence Solutions](#)

Third-party documentation

The following documentation provides information about partner technologies:

- [NVIDIA Spectrum-X](#)
- [BE Networks - Intent-based Networking \(IBN\) Orchestration Platform](#)